Week 8: Database Relationship Analysis Using Joins

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Tasks: 1.Display customer information along with their orders

```
SQL> SELECT C.*, O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------

       1001 Arun
arun@gmail.com                                     9876543210       1001
27-SEP-26        67500 Pending           1001

       1002 Priya
priya@gmail.com                                    9876543211       1002
27-SEP-26        25000 Shipped           1002

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------


       1003 Kumar
kumar@gmail.com                                    9876543212       1003
27-SEP-26         1500 Delivered         1003

       1004 Divya
divya@gmail.com                                    9876543213       1004

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------
27-SEP-26         4200 Pending           1004

       1005 Rahul
rahul@gmail.com                                    9876543214       1005
27-SEP-26         8000 Delivered         1005
```

2.DISPLAY ORDER DETAILS WITH PAYMENT INFORMATION.

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, P.Payment_ID, P.Payment_Date, P.Payment_Mode, P.Payment_Status, P.Transaction_Amount FROM Orders O INNER JOI
N Payment P ON O.Order_ID = P.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------

      1001 27-SEP-26        67500 Pending             1 27-SEP-26
UPI                  Success                67500

      1001 27-SEP-26        67500 Pending             6 29-SEP-26
Card                 Failed                  1000

      1002 27-SEP-26        25000 Shipped             2 27-SEP-26
Card                 Success                25000


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------

      1002 27-SEP-26        25000 Shipped             7 30-SEP-26
UPI                  Pending                 5000

      1003 27-SEP-26         1500 Delivered           3 28-SEP-26
Cash                 Success                1500

      1003 27-SEP-26         1500 Delivered           8 30-SEP-26
Net Banking          Failed                 2000


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------

      1004 27-SEP-26         4200 Pending             4 28-SEP-26
Net Banking          Pending                4200

      1005 27-SEP-26         8000 Delivered           5 29-SEP-26
UPI                  Success                8000
```

3.RETRIEVE PRODUCT PURCHASED BY CUSTOOMERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID, P.Product_ID, P.Product_Name, P.Price FROM Customer C INNER JOIN Orders O ON C.Customer_ID = O.Customer_ID INNER JOIN Orde
r_Details OD ON O.Order_ID = OD.Order_ID INNER JOIN Product P ON OD.Product_ID = P.Product_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
       1001 Arun                                                    1001
P1001      Laptops                  67500

       1002 Priya                                                   1002
P1002      Mobile                   25000

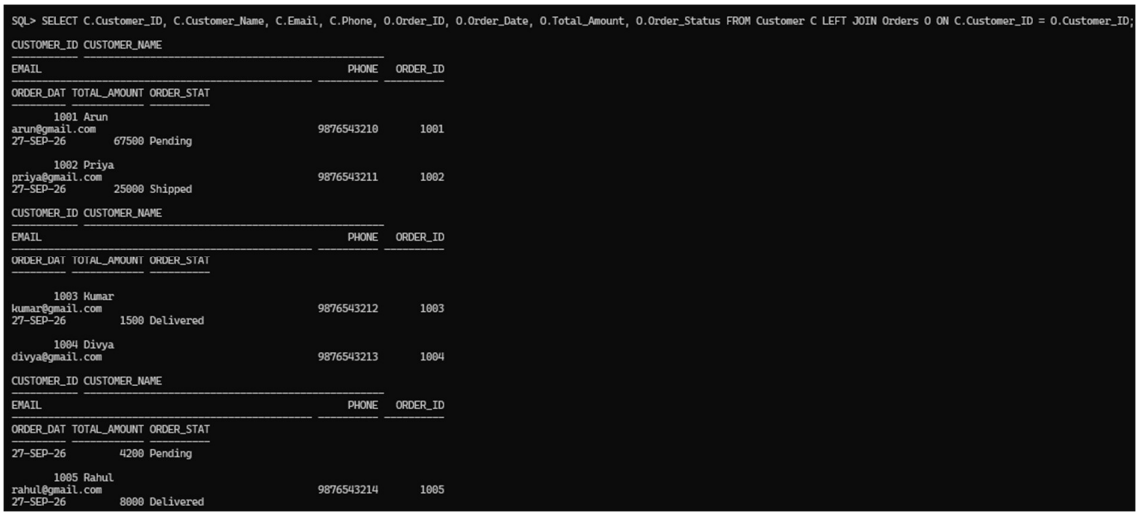
       1003 Kumar                                                   1003
P1003      TShirt                     800


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
       1004 Divya                                                   1004
P1004      Shoes                     1500
```

IMPLEMENT LEFT JOIN

TASKS:

1.DISPLAY ALL CUSTOMERS INCLUDING CUSTOMERS WHO HAVE NOT PLACED ORDERS.



2.FIND CUSTOMERS WITHOUT ANY PURCHASES.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, C.Email, C.Phone FROM Customer C LEFT JOIN Orders O
ON C.Customer_ID = O.Customer_ID WHERE O.Order_ID IS NULL ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                    PHONE
-------------------------------------------------- ----------
       1006 Rohan
rohan@gmail.com                                    9876543215
```

3.DISPLAY ALL PRODUCTS INCLUDING PRODUCTS WITH NO SALES.

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Price FROM Product P LEFT JOIN Order_Details OD ON P
.Product_ID = OD.Product_ID WHERE OD.Product_ID IS NULL ORDER BY P.Product_ID;

PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
P1006      Mouse                       500
```

IMPLEMENT RIGHT JOIN

TASK:

1.DISPLAY ALL ORDERS WITH CUSTOMERS INFORMATION

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, C.Customer_ID, C.Customer_Name, C.Email FROM Customer C RIGHT JOIN Orders O ON C.Customer_ID = O.Customer_ID
 ORDER BY O.Order_Date;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
      1001 27-SEP-26        67500 Pending           1001
Arun
arun@gmail.com

      1002 27-SEP-26        25000 Shipped           1002
Priya
priya@gmail.com

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------

      1003 27-SEP-26         1500 Delivered         1003
Kumar
kumar@gmail.com

      1004 27-SEP-26         4200 Pending           1004
Divya

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
divya@gmail.com

      1005 27-SEP-26         8000 Delivered         1005
Rahul
rahul@gmail.com
```

2.FIND ORDERS WHERE CUSTOMERS DETAILS ARE MISSING.

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, O.Customer_ID FROM Customer C RIGHT JOIN Or
ders O ON C.Customer_ID = O.Customer_ID WHERE C.Customer_ID IS NULL;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1006 25-OCT-26         1500 Pending           9999
```

3.DISPLAY ALL PAYMENT RECORDS WITH ORDER DETAILS.

```
SQL> SELECT P.Payment_ID, P.Order_ID, P.Payment_Date, P.Payment_Mode, P.Payment_Status, P.Transaction_Amount, O.Order_Date, O.Total_Amount, O.Order_Status FROM Payment P INNER JOIN Orders O ON P.Order_ID = O.
Order_ID ORDER BY P.Payment_ID;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE        PAYMENT_ST
---------- ---------- --------- -------------------- ---------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         1       1001 27-SEP-26 UPI                  Success
             67500 27-SEP-26        67500 Pending

         2       1002 27-SEP-26 Card                 Success
             25000 27-SEP-26        25000 Shipped

         3       1003 28-SEP-26 Cash                 Success
              1500 27-SEP-26         1500 Delivered


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE        PAYMENT_ST
---------- ---------- --------- -------------------- ---------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         4       1004 28-SEP-26 Net Banking          Pending
              4200 27-SEP-26         4200 Pending

         5       1005 29-SEP-26 UPI                  Success
              8000 27-SEP-26         8000 Delivered

         6       1001 29-SEP-26 Card                 Failed
              1000 27-SEP-26        67500 Pending


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE        PAYMENT_ST
---------- ---------- --------- -------------------- ---------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         7       1002 30-SEP-26 UPI                  Pending
              5000 27-SEP-26        25000 Shipped

         8       1003 30-SEP-26 Net Banking          Failed
              2000 27-SEP-26         1500 Delivered

8 rows selected.
```

5.RETRIEVE COMPLETE ORDERS DETAILS

TASK:

GENERAATE A COMPLETE ORDER REPORT BY COMBINING:

- CUSTOMER NAME

- PRODUCT NAME
- QUANTITY
- ORDER DATE
- TOTAL AMOUNT
- PAYMENT STATUS

```
SQL> SELECT C.Customer_Name, P.Product_Name, OD.Quantity, O.Order_Date, O.Total_Amount, PY.Payment_Status FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer_ID JOIN Order_Details OD ON O.Order_ID = O
D.Order_ID JOIN Product P ON OD.Product_ID = P.Product_ID LEFT JOIN Payment PY ON O.Order_ID = PY.Order_ID ORDER BY O.Order_ID;

CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Arun                                               Laptops
         1 27-SEP-26        67500 Success

Arun                                               Laptops
         1 27-SEP-26        67500 Failed

Priya                                              Mobile
         1 27-SEP-26        25000 Success

CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Priya                                              Mobile
         1 27-SEP-26        25000 Pending

Kumar                                              TShirt
         2 27-SEP-26         1500 Success

Kumar                                              TShirt
         2 27-SEP-26         1500 Failed

CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Divya                                              Shoes
         2 27-SEP-26         4200 Pending

Rahul                                              Keyboard
         1 27-SEP-26         8000 Success

8 rows selected.
```

6.DISPLAY CUSTOMER PURCHASE HISTORY

TASKS:

1.ALL PRODUCTS PURCHASED BY A CUSTOMER.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, P.Product_ID, P.Product_Name, OD.Quantity, OD.Price FROM Customer C JOIN Orders O ON C.Custome
r_ID = O.Customer_ID JOIN Order_Details OD ON O.Order_ID = OD.Order_ID JOIN Product P ON OD.Product_ID = P.Product_ID ORDER BY C.Customer_
ID;

CUSTOMER_ID CUSTOMER_NAME                                      PRODUCT_ID
----------- -------------------------------------------------- ----------
PRODUCT_NAME           QUANTITY      PRICE
-------------------- ---------- ----------
       1001 Arun                                               P1001
Laptops                       1      67500

       1002 Priya                                              P1002
Mobile                        1      25000

       1003 Kumar                                              P1003
TShirt                        2        750


CUSTOMER_ID CUSTOMER_NAME                                      PRODUCT_ID
----------- -------------------------------------------------- ----------
PRODUCT_NAME           QUANTITY      PRICE
-------------------- ---------- ----------
       1004 Divya                                              P1004
Shoes                         2       2100

       1005 Rahul                                              P1005
Keyboard                      1       8000
```

2.TOTAL AMOUNT SPENT BY EACH CUSTOMERS.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Spent FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer
_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_SPENT
----------- -------------------------------------------------- -----------
       1001 Arun                                                     67500
       1002 Priya                                                    25000
       1003 Kumar                                                     1500
       1004 Divya                                                     4200
       1005 Rahul                                                     8000
```

3.NUMBER OF ORDERS PLACED BY EACH CUSTOMER

```
SQL> SELECT C.Customer_ID, C.Customer_Name, COUNT(O.Order_ID) AS Total_Orders FROM Customer C LEFT JOIN Orders O ON C.Customer_ID = O.Cust
omer_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
       1001 Arun                                                          1
       1002 Priya                                                         1
       1003 Kumar                                                         1
       1004 Divya                                                         1
       1005 Rahul                                                         1
       1006 Rohan                                                         0

6 rows selected.
```

4.LATEST PURCHASE DETAILS OF CUSTOMERS.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID, O.Order_Date AS Latest_Purchase_Date, O.Total_Amount, O.Order_Status FROM Customer
 C JOIN Orders O ON C.Customer_ID = O.Customer_ID WHERE O.Order_Date = (SELECT MAX(O2.Order_Date) FROM Orders O2 WHERE O2.Customer_ID = C.
Customer_ID) ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
LATEST_PU TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1001 Arun                                                    1001
27-SEP-26        67500 Pending

       1002 Priya                                                   1002
27-SEP-26        25000 Shipped

       1003 Kumar                                                   1003
27-SEP-26         1500 Delivered


CUSTOMER_ID CUSTOMER_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
LATEST_PU TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1004 Divya                                                   1004
27-SEP-26         4200 Pending

       1005 Rahul                                                   1005
27-SEP-26         8000 Delivered
```

7.GENERATE MULTI-TABLE BUSINESS REPORTS

REPORT 1:CUSTOMER ORDER REPORT

DISPLAY:

- CUSTOMER NAME
- ORDER ID
- ORDER DATE
- ORDER STATUS

```
SQL> SELECT C.Customer_Name, O.Order_ID, O.Order_Date, O.Order_Status FROM Customer C INNER JOIN Orders O ON C.Customer_ID = O.Customer_ID
 ORDER BY O.Order_Date;

CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
ORDER_STAT
----------
Arun                                                      1001 27-SEP-26
Pending

Priya                                                     1002 27-SEP-26
Shipped

Kumar                                                     1003 27-SEP-26
Delivered


CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
ORDER_STAT
----------
Divya                                                     1004 27-SEP-26
Pending

Rahul                                                     1005 27-SEP-26
Delivered
```

REPORT 2:SALES REPORT

DISPLAY:

- PROUDUCT NAME
- QUANTITY SOLD
- TOTAL REVENUE

```
SQL> SELECT P.Product_Name, SUM(OD.Quantity) AS Quantity_Sold, SUM(OD.Quantity * OD.Price) AS Total_Revenue FROM Product P INNER JOIN Orde
r_Details OD ON P.Product_ID = OD.Product_ID GROUP BY P.Product_ID, P.Product_Name ORDER BY Total_Revenue DESC;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
Laptops                          1         67500
Mobile                           1         25000
Keyboard                         1          8000
Shoes                            2          4200
TShirt                           2          1500
```

REPORT 3: PAYMENT ANALYSIS REPORT

DISPLAY:

- PAYMENT MODE
- NUMBER OF TRANSACTION
- SUCCESSFUL PAYMENTS

```
SQL> SELECT Payment_Mode, COUNT(*) AS Number_Of_Transactions, SUM(CASE WHEN Payment_Status = 'Success' THEN 1 ELSE 0 END) AS Successful_Pa
yments FROM Payment GROUP BY Payment_Mode ORDER BY Payment_Mode;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- -------------------
Card                                      2                   1
Cash                                      1                   1
Net Banking                               2                   0
UPI                                       3                   2
```

REPORT 4: CUSTOMER PURCHASE ANALYSIS

FIND:

TOP PURCHASING CUSTOMERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Purchase FROM Customer C JOIN Orders O ON C.Customer_ID = O.Custo
mer_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
       1001 Arun                                                        67500
       1002 Priya                                                       25000
       1005 Rahul                                                        8000
       1004 Divya                                                        4200
       1003 Kumar                                                        1500
```

CUSTOMERS WITH MAXIMUM ORDERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, COUNT(O.Order_ID) AS Total_Orders FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer_
ID GROUP BY C.Customer_ID, C.Customer_Name HAVING COUNT(O.Order_ID) = (SELECT MAX(Order_Count) FROM (SELECT COUNT(*) AS Order_Count FROM O
rders GROUP BY Customer_ID)) ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
       1001 Arun                                                         1
       1002 Priya                                                        1
       1003 Kumar                                                        1
       1004 Divya                                                        1
       1005 Rahul                                                        1
```

CUSTOMERS WITH HIGHEST SPENDING

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Spending FROM Customer C JOIN Orders O ON C.Customer_ID = O.Custo
mer_ID GROUP BY C.Customer_ID, C.Customer_Name HAVING SUM(O.Total_Amount) = (SELECT MAX(Total_Spending) FROM (SELECT SUM(Total_Amount) AS
Total_Spending FROM Orders GROUP BY Customer_ID));

CUSTOMER_ID CUSTOMER_NAME                                     TOTAL_SPENDING
----------- -------------------------------------------------- --------------
       1001 Arun                                                        67500
```